

Diet Success Analysis and Prediction System

Elements of Artificial Intelligence and Data Science

Group 7

Erick Gonzalo Salas Duque – 55594

Elkin Jahir Figueroa Torres – 55591

Ismail Can Bingöl – 57343

May 2026

Abstract

This project develops a simple data science pipeline to analyze and predict diet outcomes using patient, diet, nutritionist, and clinical result data. The original dataset is composed of four CSV files, which are first integrated into a single dataset and then prepared through data cleaning, missing value handling, outlier treatment, and normalization.

The project follows the required tasks: data integration, preprocessing, exploratory data analysis, pattern identification, prediction modeling, and discussion of the results. The main prediction target is the patient weight change after six months, represented by `weight_change_kg_6m`. In addition, a simple binary target is created to estimate diet success probability, where a diet is considered successful if the patient loses at least 5 kg in six months.

The results show that behavioral and physical variables, especially diet adherence, baseline weight, age, and BMI, are relevant for explaining weight change. For regression, three models are compared: Linear Regression, Decision Tree Regressor, and Random Forest Regressor. In the final test set, Linear Regression obtains the best RMSE, with a Test RMSE of 2.5266 and a Test R^2 of 0.8818, while Random Forest also performs well with a Test RMSE of 3.1535 and a Test R^2 of 0.8158. A simple Decision Tree Classifier is also used to estimate the probability of diet success.

Overall, the project shows that machine learning models can help analyze diet outcomes and identify important factors related to weight loss. However, the results should be interpreted carefully, considering the simplicity of the models, the chosen definition of success, and the characteristics of the available dataset.

1 Data Integration

The first task of the project was to integrate the four CSV files into a single unified dataset. This step was implemented in the script `src/data_integration.py`, mainly through the function `integrate_data()`. The original data sources were:

- `patients.csv`
- `diets.csv`

- `nutritionists.csv`
- `outcomes.csv`

The `outcomes.csv` file was used as the central table because it contains the clinical results of the diet programs. The remaining datasets were merged into this table using their corresponding identifiers. First, `outcomes.csv` was merged with `patients.csv` using `patient_id`. Then, the resulting dataset was merged with `diets.csv` using `diet_id`. Finally, it was merged with `nutritionists.csv` using `nutritionist_id`.

All merge operations were performed using left joins. This choice preserves all records from the outcomes table, even when some related information is missing in the patient, diet, or nutritionist tables. This is important because the outcome records represent the main observations used later for analysis and prediction.

After the integration step, the final merged dataset contained 2523 rows and 31 columns. The output was saved as `data/merged_data.csv`.

	patient_id	diet_id	nutritionist_id	age	sex	baseline_weight_kg	baseline_bmi	diet_type	mean_adherence_pct	motivation_score	weight_change_kg_6m
0	P0001	D003	N013	42.0	F	78.1	32.2	balanced	85.7	NaN	-20.7
1	P0001	D002	N014	42.0	F	78.1	32.2	balanced	100.0	NaN	-25.6
2	P0001	D009	N015	42.0	F	78.1	32.2	high_protein	88.7	NaN	-24.8
3	P0001	D007	N005	42.0	F	78.1	32.2	high_protein	91.4	NaN	-25.8
4	P0002	D007	N013	28.0	F	88.2	33.4	high_protein	89.0	0.724	-28.1
5	P0002	D003	N009	28.0	F	88.2	33.4	balanced	78.2	0.724	-26.5
6	P0002	D004	N007	28.0	F	88.2	33.4	low_carb	80.8	0.724	-30.9
7	P0003	D007	N015	33.0	F	58.9	22.9	high_protein	82.9	0.630	-18.8
8	P0003	D007	N018	33.0	F	58.9	22.9	high_protein	69.5	0.630	-14.4
9	P0003	D004	N020	33.0	F	58.9	22.9	low_carb	75.9	0.630	-20.3
10	P0003	D005	N017	33.0	F	58.9	22.9	low_carb	81.9	0.630	-23.0
11	P0004	D003	N008	53.0	F	89.4	29.9	balanced	46.3	0.393	-10.6

Figure 1: Example rows from the merged dataset after data integration.

Figure 1 shows a sample of the merged dataset after the integration step. The table contains identifiers for patients, diets, and nutritionists, together with patient characteristics such as age, sex, baseline weight and BMI, diet type, adherence percentage, motivation score, and the target variable `weight_change_kg_6m`.

This integration step was necessary because the later stages of the project require information from all four sources at the same time. For example, predicting weight change requires patient information, diet information, nutritionist information, and the final outcome to be available in one unified table.

2 Data Cleaning and Preprocessing

The second task of the project was to clean and preprocess the merged dataset. This step was implemented in the script `src/preprocessing.py`, mainly through the function `clean_data()`. The function receives the integrated dataset, `data/merged_data.csv`, and prepares it for the following stages of the project.

First, duplicate rows were removed from the dataset. In this project, 3 duplicate rows were found and removed. This avoids repeated observations affecting the exploratory analysis and the training of the prediction models.

After that, redundant and identifier columns were removed. The removed columns were:

- `patient_id`

- `diet_id`
- `nutritionist_id`
- `bmi_redundant`
- `experience_years`

The identifier columns were useful during the data integration step, but they were removed before modeling because they do not represent meaningful predictive features. The column `bmi_redundant` was removed because it duplicated BMI-related information already present in the dataset.

Missing values were handled in two different ways. Rows with missing values in the target variable, `weight_change_kg_6m`, were removed because they cannot be used for supervised learning. In total, 151 rows with missing target values were dropped. For the remaining missing values, numerical columns were filled using the median, while categorical columns were filled using the mode.

Outliers were handled by capping numerical values at three standard deviations from the mean. This reduces the influence of extreme values while keeping the corresponding records in the dataset. The target variable was not capped, so the original weight-change values were preserved.

Finally, numerical features were normalized using `StandardScaler`. The target variable `weight_change_kg_6m` was excluded from normalization because it is the value to be predicted. Normalization is especially important for methods that depend on feature scale, such as clustering.

The final cleaned and normalized dataset was saved as `data/cleaned_data.csv`.

Step	Action
Duplicate removal	Removed repeated rows
Column removal	Removed IDs and redundant columns
Missing target values	Removed rows without <code>weight_change_kg_6m</code>
Missing feature values	Filled numerical values with median and categorical values with mode
Outlier treatment	Capped numerical features at 3 standard deviations
Normalization	Applied <code>StandardScaler</code> to numerical features

Table 1: Summary of the preprocessing steps.

3 Exploratory Data Analysis

The third task of the project was to perform exploratory data analysis on the cleaned dataset. This step was implemented in the script `src/eda_analysis.py`, mainly through the function `run_eda()`. The goal of this step was to understand the main characteristics of the data, identify possible relationships between variables, and observe how different patient and diet features are related to the final weight change.

First, summary statistics were generated for the numerical variables using the `describe()` function from Pandas. These statistics were saved in `eda_results/summary_statistics.csv`. This output includes measures such as mean, standard deviation, minimum, maximum, and quartiles, which help provide a general overview of the cleaned dataset.

After that, a correlation matrix was created using the numerical variables. This matrix helps identify linear relationships between variables such as age, baseline BMI, adherence percentage, motivation score, and weight change after six months.

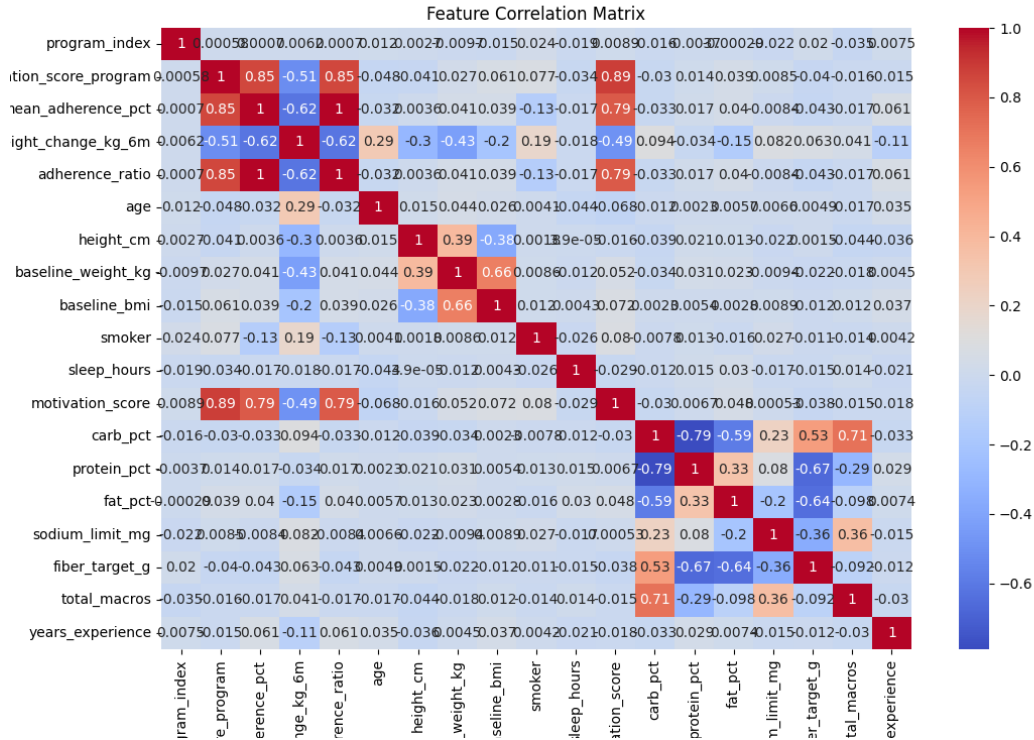


Figure 2: Correlation matrix of the numerical variables.

Figure 2 shows the correlation between the numerical features in the dataset. This analysis is useful for understanding which variables may be more related to the target variable **weight_change_kg_6m**. In general, behavioral variables such as adherence percentage and motivation are important to consider because they describe how well the patient followed the diet program.

The analysis also compared weight change according to categorical variables. First, the relation between patient sex and weight change was analyzed using a boxplot.

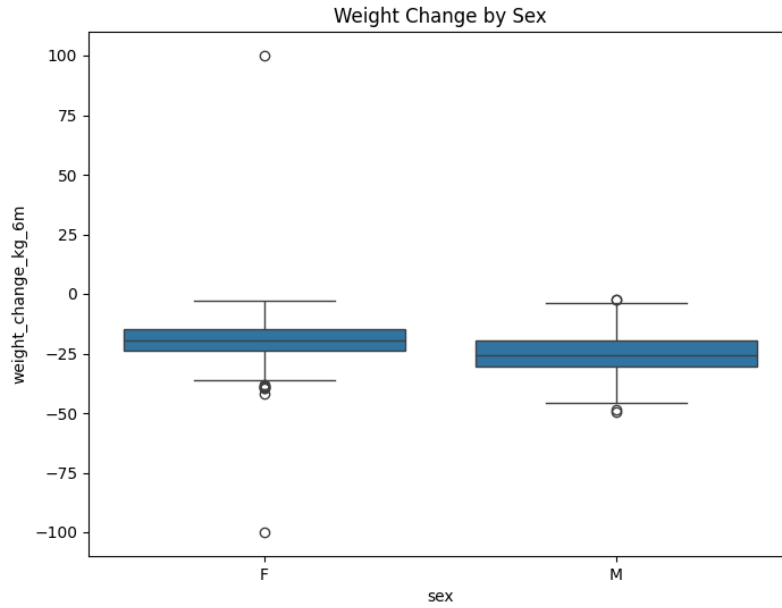


Figure 3: Weight change distribution by patient sex.

Figure 3 shows the distribution of weight change for male and female patients. This type of visualization helps identify whether the final result of the diet differs strongly between groups. In this project, sex was considered as one of the possible explanatory variables, but it was not analyzed alone because weight loss depends on several factors at the same time.

Next, the relation between diet type and weight change was analyzed. This is important because different diet types may have different effects on patient outcomes.

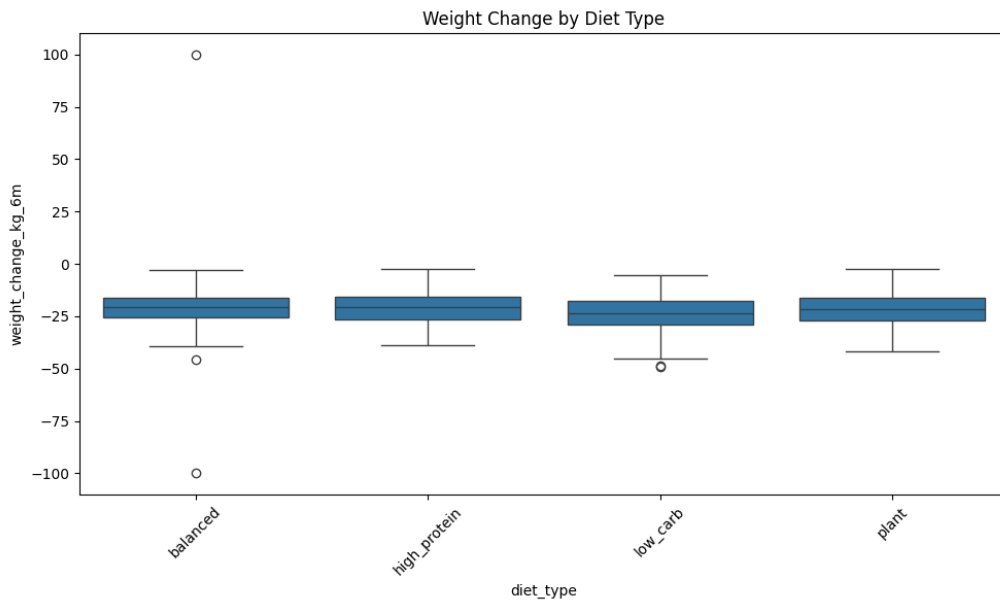


Figure 4: Weight change distribution by diet type.

Figure 4 compares the distribution of weight change across different diet types. This

allows us to observe whether some diet categories are associated with better average weight loss. However, these results should be interpreted carefully, because diet success also depends on patient characteristics, motivation, adherence, and the nutritionist profile.

Finally, an outlier visualization was created for the numerical variables. This was useful to inspect whether some variables contained extreme values after preprocessing.

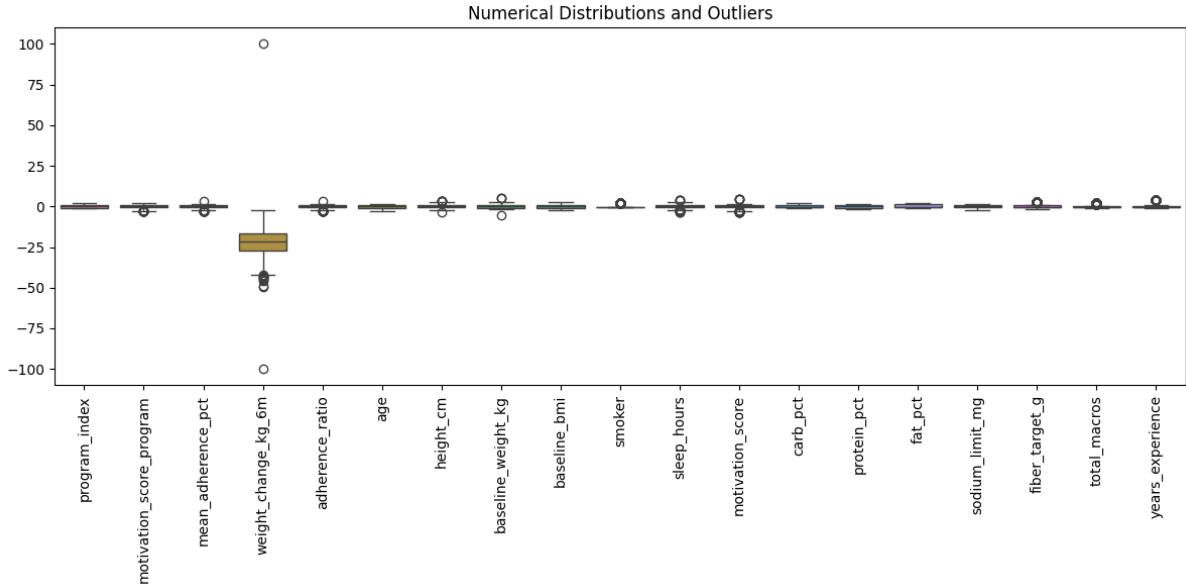


Figure 5: Boxplot visualization of numerical variables for outlier inspection.

Figure 5 shows the distribution of the numerical variables after preprocessing. Since outliers were handled during the cleaning step using capping at three standard deviations, this plot helps confirm that extreme values were reduced while keeping the records in the dataset.

Overall, the exploratory data analysis provided a first understanding of the dataset and helped guide the next stages of the project. The generated plots showed how numerical and categorical variables relate to weight change, and they also supported the later steps of pattern identification and prediction modeling.

4 Pattern Identification and Clustering

The fourth task of the project was to identify patterns in the data using unsupervised learning. This step was implemented in the script `src/clustering.py`, mainly through the function `run_clustering()`. The goal was to group patients with similar profiles and then compare the average weight change of each group.

For this task, the K-Means algorithm was used. K-Means is a simple and widely used clustering method that separates the data into a fixed number of groups. In this project, the number of clusters was set to four. This choice keeps the analysis simple and allows the groups to be compared easily.

The clustering model used the following features:

- age
- baseline_bmi

- `motivation_score`
- `mean_adherence_pct`

These variables were selected because they represent basic patient characteristics and behavioral information. Before applying K-Means, the selected features were standardized using `StandardScaler`. This was necessary because K-Means is sensitive to the scale of the variables.

After fitting the model, each patient was assigned to one of the four clusters. The average values of the selected features and the average `weight_change_kg_6m` were then calculated for each cluster. The results were saved in `clustering_results/cluster_averages.csv`. The full dataset with the assigned cluster labels was also saved in `clustering_results/clustered_data.csv`.

To evaluate the clustering result, the silhouette score was calculated. The obtained silhouette score was:

Silhouette score: 0.2450

This score indicates that the clusters are moderately separated. Since the dataset contains real-world style patient information, very clear cluster separation is not expected. However, the clustering is still useful for identifying general patient groups and comparing their average outcomes.

Cluster	Number of Patients	Average Weight Change (kg)
0	494	-25.80
1	600	-21.82
2	555	-14.57
3	720	-24.85

Table 2: Average weight change by cluster.

Table 2 shows that the clusters have different average weight changes. Cluster 2 had the weakest average result, with an average weight change of -14.57 kg, while Cluster 0 had the strongest average result, with an average weight change of -25.80 kg. This suggests that patient profile and adherence-related features can be useful for identifying groups with different diet outcomes.

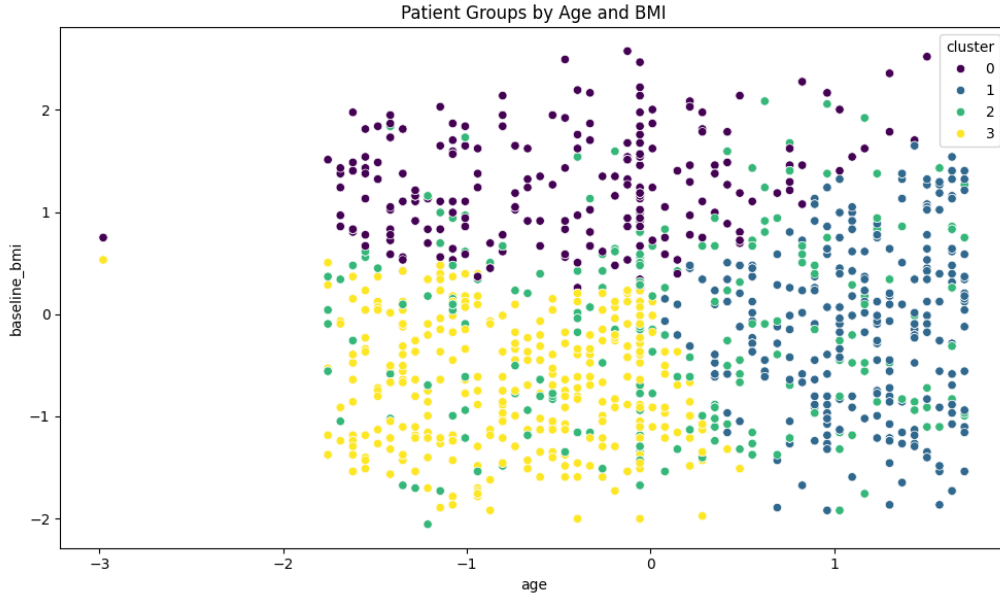


Figure 6: Patient clusters visualized by age and baseline BMI.

Figure 6 shows the four clusters using age and baseline BMI as the visualization axes. Although the model used four features, this two-dimensional plot helps provide a simple visual interpretation of the patient groups.

In addition to clustering, two specific patterns mentioned in the project statement were tested in the script `src/pattern_tests.py`. The first pattern checked whether patients younger than 50 years old lost more weight during colder months. In this dataset, the average weight change for younger patients was -23.95 kg in colder months and -23.32 kg in warmer months, so this pattern was confirmed.

The second pattern checked whether stricter nutritionists had more male patients. The percentage of male patients with strict nutritionists was 48.28%, while the percentage for other nutritionist approaches was 45.39%. Therefore, this pattern was also confirmed in the dataset.

Overall, this step showed that unsupervised learning can help identify patient groups with different average diet outcomes. The additional pattern tests also helped answer specific questions from the project statement in a simple and interpretable way.

5 Prediction Modeling

The fifth task of the project was to apply supervised learning models to predict diet outcomes. This step was implemented in the script `src/modeling.py`, mainly through the function `run_modeling()`. The main target variable was `weight_change_kg_6m`, which represents the patient weight change after six months.

Before training the models, categorical variables were converted into numerical variables using one-hot encoding. The column `record_created_at` was transformed into a simpler month feature, and identifier columns such as `program_id` were removed because they do not represent meaningful predictive information.

The dataset was divided according to the recommendation in the project statement:

- 50% for training

- 30% for validation
- 10% for testing
- 10% left unused to reduce overfitting

The resulting split contained 1184 training samples, 711 validation samples, and 237 testing samples.

Three regression models were trained and compared:

- **Linear Regression:** used as a simple baseline model.
- **Decision Tree Regressor:** used because it is interpretable and recommended in the project statement.
- **Random Forest Regressor:** used as an ensemble model to improve prediction performance.

For the Decision Tree Regressor, a simple hyperparameter selection was performed using the validation set. The tested values for `max_depth` were 3, 5, and 7. The best validation result was obtained with `max_depth = 5`.

The models were evaluated using RMSE, MAE, and R^2 . RMSE and MAE measure the prediction error in kilograms, while R^2 measures how much of the variance in the target variable is explained by the model.

Model	Test RMSE	Test MAE	Test R^2
Linear Regression	2.5266	1.9414	0.8818
Decision Tree depth=5	8.4780	3.4797	-0.3313
Random Forest	3.1535	2.1243	0.8158

Table 3: Regression model performance on the test set.

Table 3 shows that Linear Regression achieved the best result on the test set, with the lowest RMSE and the highest R^2 . Random Forest also performed well, but slightly worse than Linear Regression in this experiment. The Decision Tree Regressor performed worse, especially on the test set, which suggests that the tree did not generalize well to unseen data.

Although the Decision Tree result was weaker, it is still useful because it provides an interpretable model structure. This is important for explaining the prediction process to a non-technical audience.

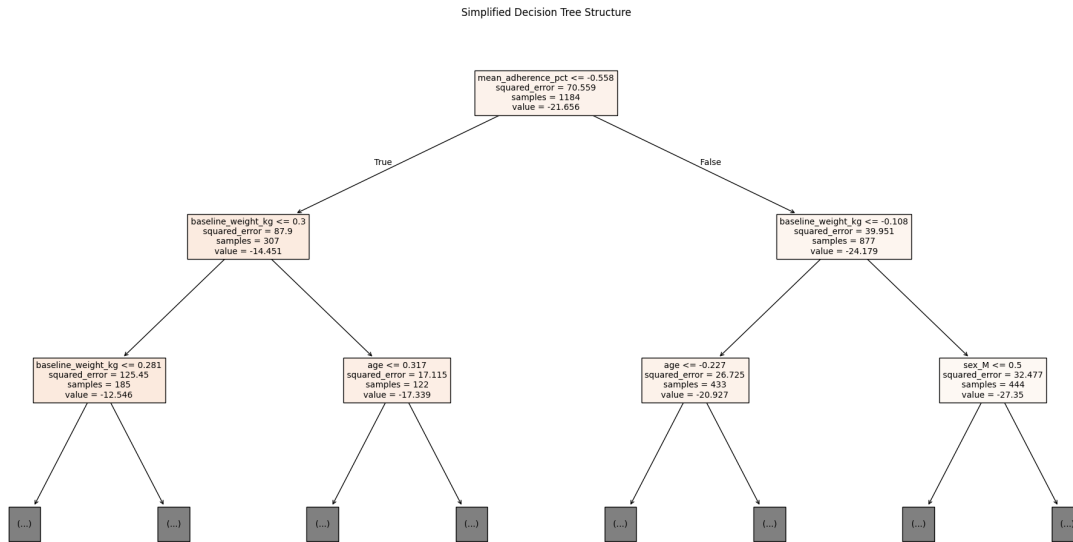


Figure 7: Simplified visualization of the Decision Tree Regressor.

Figure 7 shows the simplified structure of the Decision Tree Regressor. This visualization helps understand which variables are used near the top of the tree and how the model separates patients into different prediction groups.

To better understand which variables were important for prediction, feature importance was extracted from the Random Forest model.

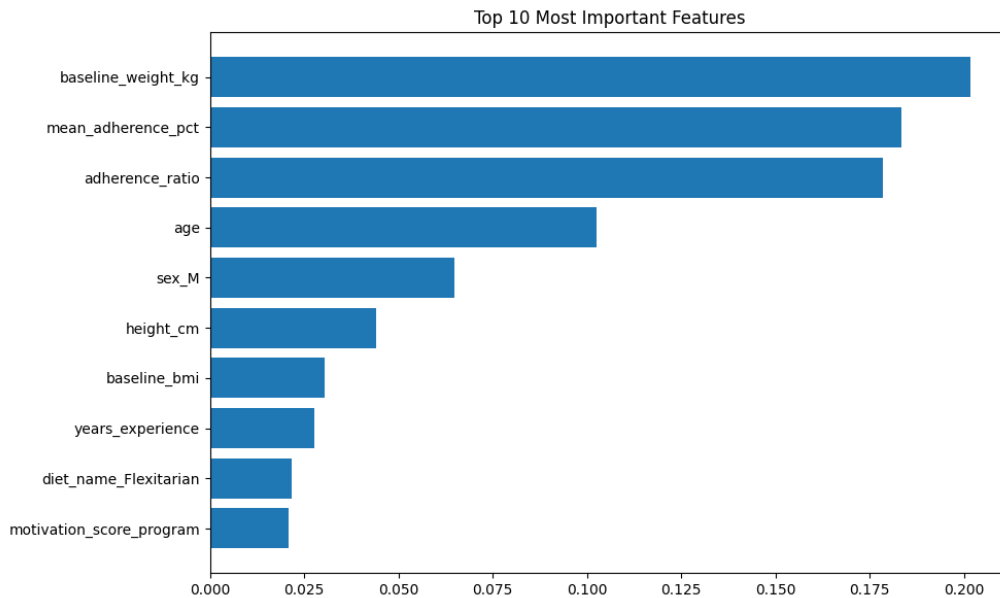


Figure 8: Top 10 most important features according to the Random Forest model.

Figure 8 shows that variables such as baseline weight, mean adherence percentage, adherence ratio, age, sex, height, and baseline BMI were among the most important features. This result is consistent with the idea that both physical characteristics and behavioral variables influence weight loss.

In addition to predicting the amount of weight change, a simple classification model was created to estimate the probability of diet success. In this project, a diet was considered successful if the patient lost at least 5 kg in six months:

$$\text{success} = \begin{cases} 1, & \text{if } \text{weight_change_kg_6m} \leq -5 \\ 0, & \text{otherwise} \end{cases}$$

A `DecisionTreeClassifier` with `max_depth = 5` was trained using the same training, validation, and test split. The model outputs a probability of success for each patient in the test set.

Model	Test Accuracy	Test F1
Decision Tree Classifier	0.9873	0.9936

Table 4: Success classification model performance on the test set.

Table 4 shows that the simple Decision Tree Classifier achieved high accuracy and F1 score. However, this result should be interpreted carefully because the definition of success may create an imbalanced target variable. Therefore, the classification model is useful as a simple demonstration of success probability, but the regression models provide more detailed information about the expected weight change.

Overall, the prediction modeling step showed that diet outcomes can be estimated using patient, diet, and behavior-related features. Linear Regression achieved the strongest test performance in this experiment, while Random Forest helped identify the most important predictive variables.

6 Critical Discussion of Results

The results obtained in this project show that the dataset contains useful information for understanding and predicting diet outcomes. However, the models and patterns must be interpreted carefully because the project uses a simple pipeline and a limited set of assumptions.

In the exploratory data analysis, variables related to patient behavior and physical condition appeared to be important. For example, adherence percentage, baseline weight, age, and BMI were relevant variables in the later prediction models. This is reasonable because weight loss is influenced not only by the type of diet, but also by how consistently the patient follows the plan and by the initial physical condition of the patient.

The clustering step helped identify groups of patients with different average outcomes. The silhouette score was 0.2450, which indicates that the clusters are not perfectly separated. This means that the groups should not be interpreted as very strict or clearly isolated patient types. Instead, they should be seen as approximate groups that help compare general tendencies. Even with this limitation, the clusters showed different average weight changes, suggesting that unsupervised learning can be useful for discovering general patient profiles.

For prediction modeling, Linear Regression achieved the best result on the test set, with a Test RMSE of 2.5266 and a Test R^2 of 0.8818. This means that, in this experiment, a simpler linear model generalized better than the more complex models. One possible explanation is that the relationship between the selected variables and weight change

is relatively regular in this dataset. Another possibility is that the dataset contains strong linear patterns, especially in variables related to adherence and baseline physical measurements.

The Decision Tree Regressor was included because it is easy to interpret and recommended for this type of project. However, it performed poorly on the test set, with a negative Test R^2 . This suggests that the tree did not generalize well to unseen data. Even though a small validation search was used to choose the best tree depth, the model may still have learned patterns that were too specific to the training data. For this reason, the Decision Tree is more useful here as an explanatory model than as the best predictive model.

The Random Forest Regressor performed better than the Decision Tree and also provided useful feature importance values. Although it did not outperform Linear Regression on the test set, it helped identify which variables were most relevant for prediction. According to the feature importance results, variables such as baseline weight, adherence percentage, adherence ratio, age, sex, height, and baseline BMI had strong influence on the predictions. This supports the idea that both physical characteristics and behavioral variables are important for diet outcomes.

The success probability model was implemented as a simple classification task. A diet was considered successful if the patient lost at least 5 kg in six months. The Decision Tree Classifier achieved high accuracy and F1 score on the test set. However, this result should be interpreted carefully. If most patients in the dataset are classified as successful according to this threshold, then high accuracy may be easier to obtain and may not fully represent the quality of the model. Therefore, this classifier is useful as a simple example of success probability, but the regression models give a more detailed view because they predict the amount of weight change directly.

Overall, the project results suggest that machine learning can support the analysis of diet outcomes, especially by identifying important variables and estimating expected weight change. However, the models should not be used as medical decision tools without further validation. In a real-world context, additional clinical variables, external validation, and expert review would be necessary before applying these predictions to patient treatment decisions.

7 Conclusion

This project implemented a complete and simple data science pipeline for analyzing and predicting diet outcomes. The work started with the integration of four CSV files containing patient, diet, nutritionist, and outcome information. After integration, the dataset was cleaned, missing values were handled, outliers were treated, and numerical variables were normalized.

Exploratory data analysis was used to better understand the dataset and observe relationships between variables. The generated plots helped analyze correlations, compare weight change across sex and diet type, and inspect possible outliers. These analyses supported the following steps of clustering and prediction modeling.

In the pattern identification step, K-Means clustering was applied to group patients with similar profiles. The clusters showed different average weight changes, which suggests that patient characteristics and behavioral variables can help identify groups with different diet outcomes. In addition, two specific patterns from the project statement were tested and confirmed in the dataset.

In the prediction modeling step, Linear Regression, Decision Tree Regressor, and Random Forest Regressor were trained to predict `weight_change_kg_6m`. The best test result was obtained by Linear Regression, while Random Forest was useful for identifying the most important features. A simple Decision Tree Classifier was also added to estimate diet success probability, where success was defined as losing at least 5 kg in six months.

Overall, the project shows that artificial intelligence and data science methods can be useful for studying diet outcomes. The models helped identify relevant variables such as adherence, baseline weight, age, and BMI. However, the results should be interpreted carefully because the models are simple and the success definition is based on a fixed threshold. Future improvements could include testing more feature combinations, using cross-validation, analyzing class imbalance in more detail, and validating the models on external data.